

MICRO ASSIGNMENT

-- 1) Write an SQL query to find the total number of patients across all hospitals.

-- ANSWER

```
SELECT SUM(Patients_Count) AS TOTAL_PATIENTS
```

```
FROM hospital_data;
```

-- 2) Retrieve the average count of doctors available in each hospital.

-- ANSWER

```
SELECT Hospital_Name, COUNT(Doctors_Count) AS AVERAGE_COUNT
```

```
FROM hospital_data
```

```
GROUP BY Hospital_Name;
```

-- 3) Find the top 3 hospital departments that have the highest number of patients.

-- ANSWER

```
SELECT Department, SUM(Patients_Count) AS NUMBER_OF_PATIENTS
```

```
FROM hospital_data
```

```
GROUP BY Department
```

```
ORDER BY NUMBER_OF_PATIENTS DESC LIMIT 3;
```

-- 4) Identify the hospital that recorded the highest medical expenses.

-- ANSWER

```
SELECT DISTINCT Hospital_Name , Medical_Expenses
```

```
FROM hospital_data
```

```
ORDER BY Medical_Expenses DESC LIMIT 1;
```

-- 5) Calculate the average medical expenses per day for each hospital.

-- ANSWER

```
SELECT Hospital_Name, AVG(Medical_Expenses/NULLIF((discharge_date-  
admission_date),0)) AS AVG_DAILY_EXPENSE
```

```
FROM hospital_data
```

```
GROUP BY Hospital_Name;
```

-- 6) Find the patient with the longest stay by calculating the difference between Discharge Date and Admission Date.

```
-- ANSWER SELECT Hospital_Name, Department, admission_date, discharge_date,  
(discharge_date-admission_date) AS STAY_DAYS
```

```
FROM hospital_data
```

```
ORDER BY STAY_DAYS DESC LIMIT 1;
```

-- 7) Count the total number of patients treated in each city.

-- ANSWER

```
SELECT location, SUM(patients_count) AS PATIENTS_TREATED
```

```
FROM hospital_data
```

```
GROUP BY location;
```

-- 8) Calculate the average number of days patients spend in each department.

-- ANSWER

```
SELECT department, AVG(discharge_date-admission_date) AS AVG_DAYS
```

```
FROM hospital_data
```

```
GROUP BY department;
```

-- 9) Find the department with the least number of patients.

-- ANSWER

```
SELECT DISTINCT department, patients_count AS NUMBER_OF_PATIENTS
```

FROM hospital_data

ORDER BY NUMBER_OF_PATIENTS ASC LIMIT 1;

-- 10) Group the data by month and calculate the total medical expenses for each month.

-- ANSWER

SELECT EXTRACT(MONTH FROM admission_date) AS Month, SUM(Medical_Expenses) AS
TOTAL_EXPENSE

FROM hospital_data

GROUP BY Month

ORDER BY Month;